

Chapter 1: Introduction to epidemiology

Objectives

- Understand the definition and objectives of epidemiology
- Differentiate between descriptive and analytic epidemiology
- Differentiate between Health, disease and illness
- Differentiate between communicable and non-communicable diseases
- Recognize why communicable diseases remain major problem in developing countries
- Recognize theories of disease causation
- Recognize the difference between infection, colonization, contamination, pollution and infestation
- Define dead-end infection and latent infection
- Recognize the difference between eradication, control and elimination

Definition of Epidemiology; it is the study of the distribution and determinants of diseases in the populations, and the application of this study to control health problems.

- **Distribution;** This refers to the pattern of occurrence of disease in the community with reference to time, place and person.
- **Determinants;** This refers to the etiological or risk factors related to the disease.

Objectives of epidemiology

1. To know the distribution of the disease in the community
2. To know the magnitude of the problem
3. To identify the etiological and risk factors in the development of disease

4. To plan for the implementation of prevention and control measures
5. To eliminate or eradicate the disease
6. To evaluate the control measures
7. Ultimate objective is to promote the health and wellbeing of the people.

Difference between Health, Disease, and Illness

- Health is a state of complete physical, mental, and social well-being of an individual and not merely an absence of disease or infirmity.

Physical dimension means all the organs and systems in the body are functioning perfectly at their optimum capacity. A person is said to be mentally healthy, when he/she is having a perfect state of balance with the surrounding world, having harmonious relation with others, the intelligence, memory, learning capacity, judgment is normal, not having any internal conflicts, accepts criticism sportively, has got good self-control. An individual is said to be socially healthy, when he is accepted, respected, and loved by all in the family, by his friends, relatives, neighbors, colleagues and others.

- disease is a physiological or psychological dysfunction of the body.
- Illness is a subjective feeling of not being well

Diseases of man can be broadly classified into communicable and non – communicable diseases.

1. **Communicable** diseases can be **transmitted** from man to man (mainly) or animal to man (sometimes) by different means.

2. Non communicable diseases; are **non-transmissible** disease ex., hypertension, diabetes Miletus and cancer.

Communicable diseases remain major problem in developing countries especially rural and slums areas **due to:**

A- insanitary environment

B- improper health habits and behavior of the public

C- poor quality and underutilized health services.

Theories of disease causation

A variety of theories of disease causation have been proposed. The purpose of these theories is to facilitate the understanding of nature of the disease, which is complex.

1- Old theories:

Till the end of the 18th century, various theories were in vogue, e.g. Supernatural theory of disease (e.g. curse of God; an evil eye).

2- Germ theory:

The discoveries in microbiology at the turn of the 18th century became a turning point in the etiological concept of disease. Louis Pasteur (1860) demonstrated the presence of bacteria in the air. Robert Koch (1877) showed that anthrax was caused by bacteria. These discoveries of Pasteur and Koch confirmed the germ theory of disease. The theory concluded that the disease is caused by single agent (microorganism).

Limitations of Germ Theory

The germ theory is unable to explain:

- a. Why only some people suffer from the disease after exposure to microorganisms and not all, as with exposure to tubercle bacilli.
- b. Why some people do not suffer from the disease even though they harbor the pathogens in the body (as in healthy carriers of typhoid)

3. Epidemiologic triad;

- Agent, host, and environmental factors interrelate to produce disease in humans.
- According to this model (Fig. 1) disease occurs when the equilibrium between agent, host and environment is disturbed. Thus, this model explains that some persons do not suffer from the disease even though they harbor the pathogens because an equilibrium is established between the causative agent and the host.
- This theory mainly linked to **communicable** diseases.

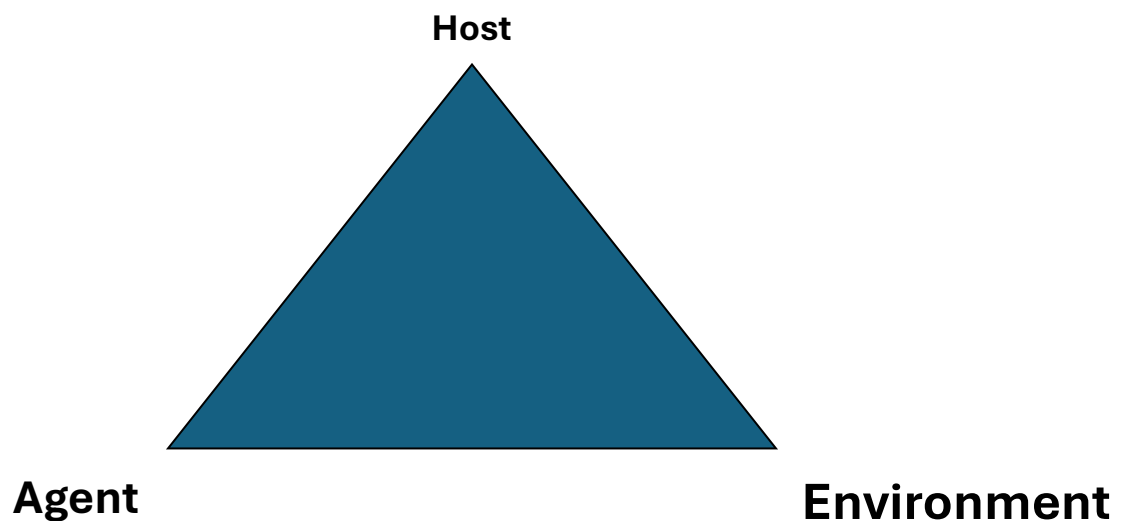


Fig. (1): Epidemiological triad

4. Theory of Multifactorial Causation (Web of Causation)

- A disease is not only caused by an organism but also predisposed by many factors contributing to its occurrence (risk factors).
- most of these risk factors are linked to life-style and human behavior.
- This theory linked mainly to modern diseases of civilization (non communicable diseases)

Important epidemiological terms

1. Infection:

- This refers to the entry and development or multiplication of an infectious agent in the human (or, animal) body, with an implied response (e.g. immunological response) on the part of the human or animal.

2. . Colonization:

- presence of infectious agent in the human body but without any evidence of specific host immune responses to the agent.

3. Contamination:

- Refers to an infectious agent being present in inanimate articles like food, water. This term is also used to denote presence of infectious agent on skin surface, particularly on hands.

4. Pollution:

- Refers to presence of either infectious agent or mechanical agents (as sound), usually in the general environment, air or water (e.g. sound pollution, water pollution, air pollution).

5. Infestation:-

- The presence and development of insect vectors on the body or linen

6. **Dead-End Infection:**

- A state when an infectious disease, which is usually “communicable”, cannot be transmitted any further between human beings or from humans to animals or vice versa, for various agent, host and environmental reasons. Examples are Rabies and Tetanus

7. **Latent Infection :**

- This refers to a state when the infectious agent lies “dormant” within the host body, without any clinical manifestations but does not come out of the human body (thus it is different from subclinical infection).
- After a period of time, under certain circumstances, the agent which had been lying dormant, reactivates (e.g. reactivation Tuberculosis).

8. **Eradication :**

- The term refers to a complete cessation of transmission of the infectious agent.
- The infectious agent as well as the disease has also been completely reduced to zero.
- Small pox is the only example wherein eradication has been achieved.

9. **Control :**

- Reducing the transmission of a disease to a level when it no longer remains a “public health problem”.

10. **Elimination :**

- Elimination implies either a ‘regional eradication’ (say from a country or continent), or else reduction of disease to zero without total removal of infectious agent.

Infectiousness, Infectivity, Pathogenicity and Virulence

- ✓ Infectiousness : this is the relative ease of agent transmission to the host.
- ✓ Infectivity : This is the ability of the agent to enter, survive and multiply in the host.
- ✓ Pathogenicity : It is the ability of the agent to produce manifest disease out of those who have been infected.
- ✓ Virulence : It is the ability of the agent to produce severe disease (fatal).

Exercise

1. Presence of an infectious agent in an inanimate article or on skin surface, particularly hands, is called

- (a) pollution (b) contamination (c) infection (d) infestation

2. The presence and development of insect vectors on the body or linen e.g. louse is known as

- (a) Infection (b) Infestation (c) Infectiousness (d) Infectivity